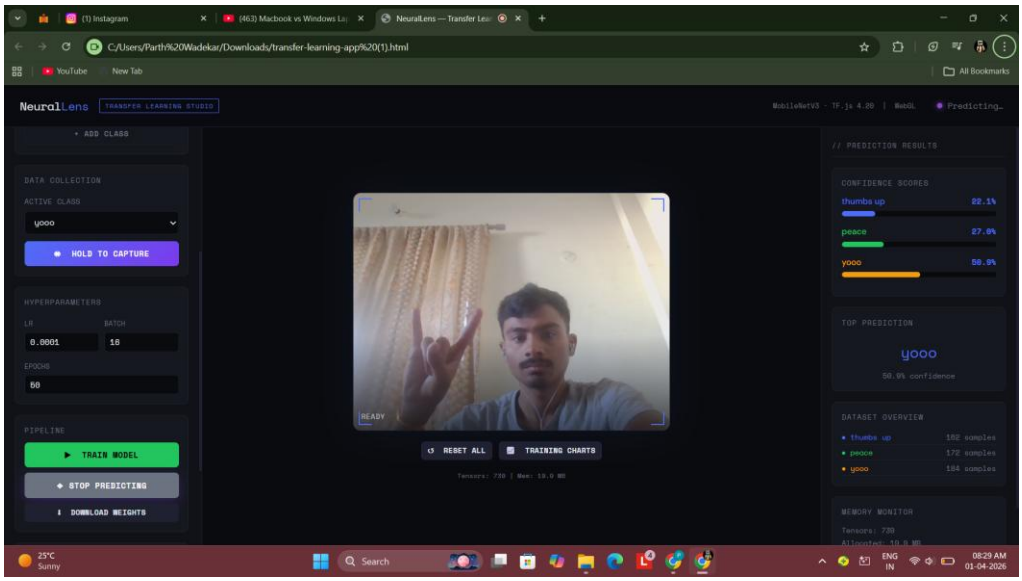
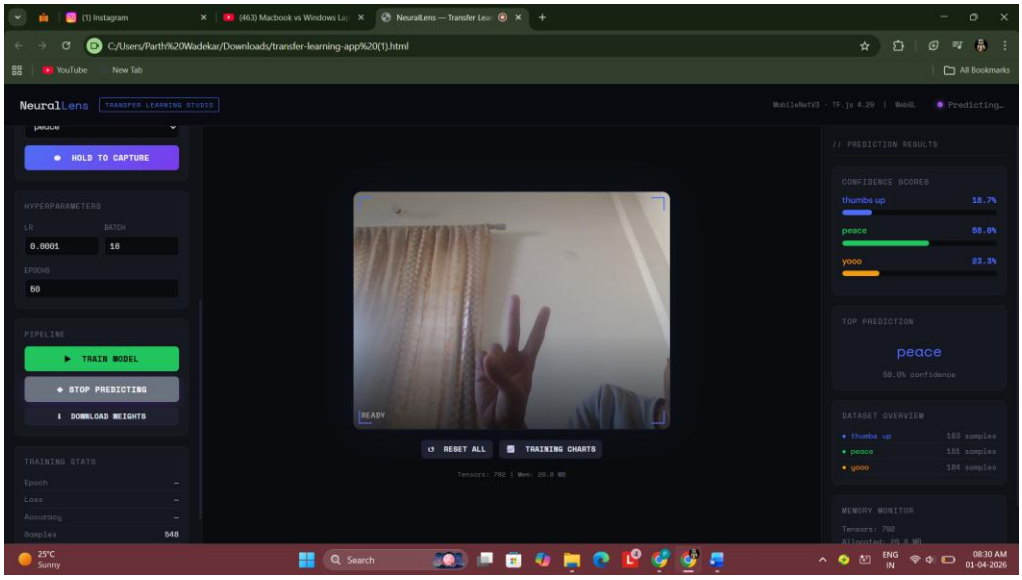
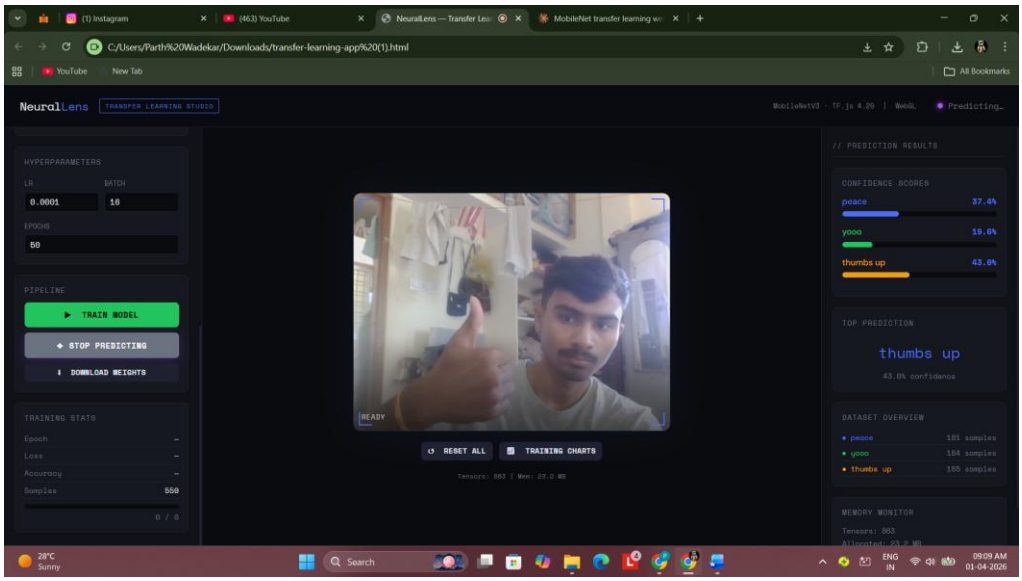
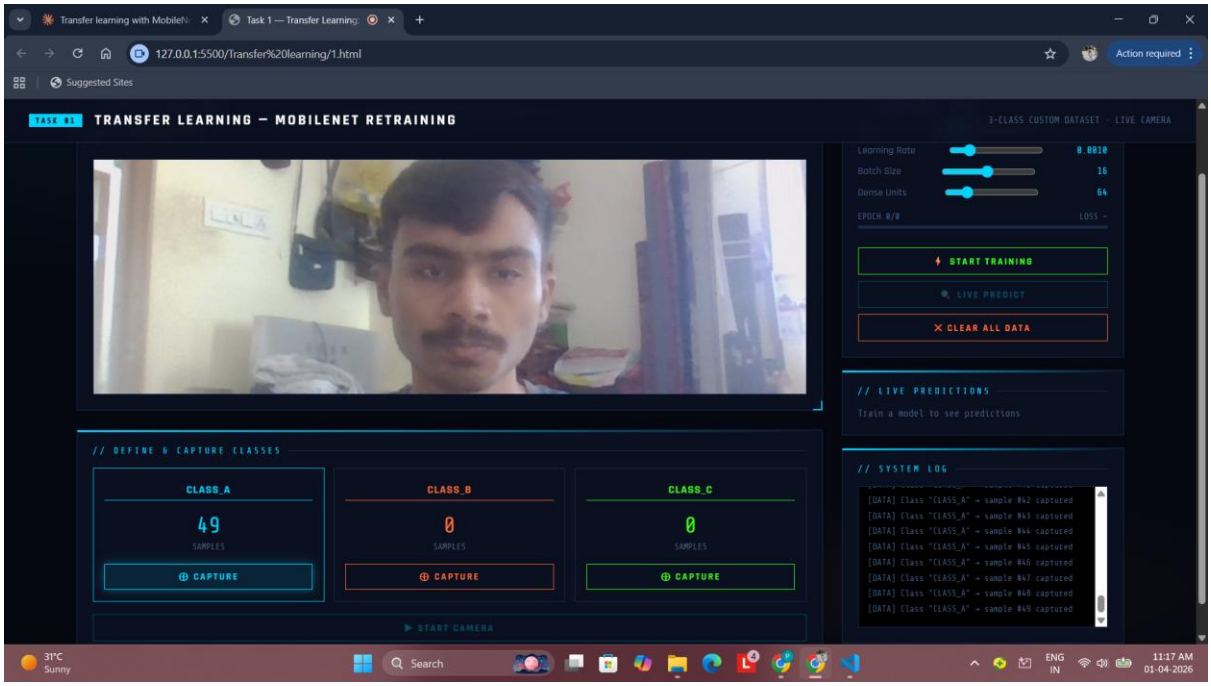
Practical 8

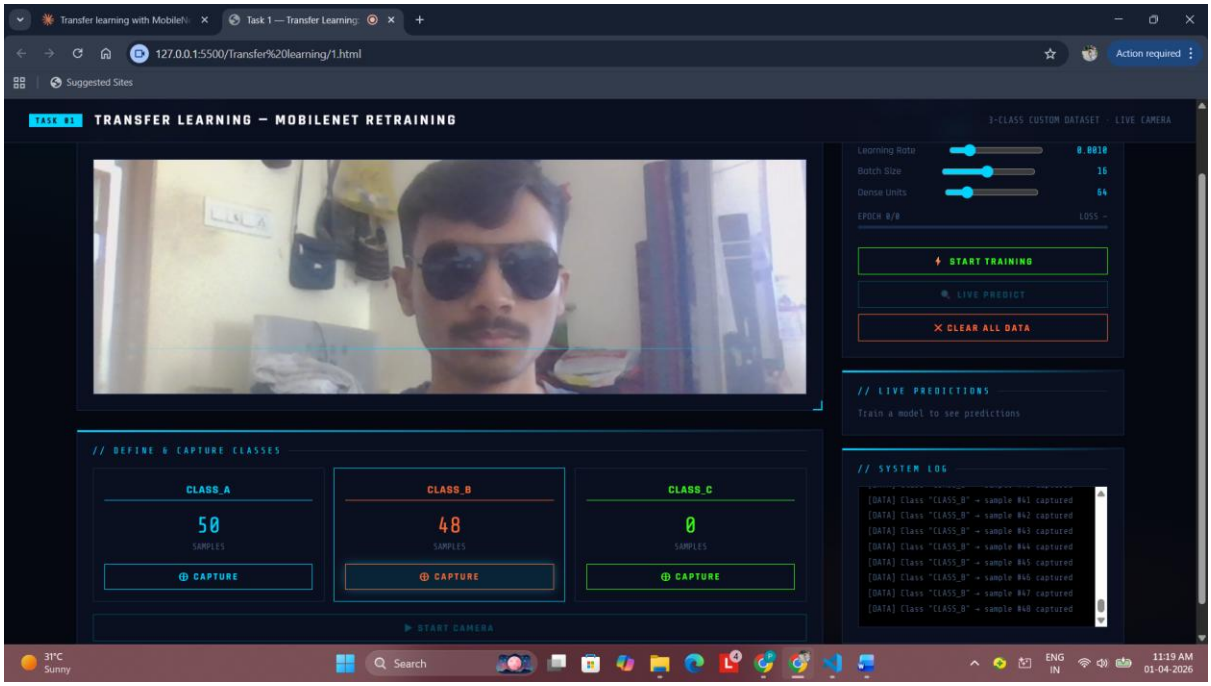

NeuralLens
TRANSFER LEARNING STUDIO
+ ADD CLASS
DATA COLLECTION
ACTIVE CLASS
yooo
HOLD TO CAPTURE
HYPERPARAMETERS
LR
BATCH
0.0001
16
EPOCHS
50
PIPELINE
TRAIN MODEL
STOP PREDICTING
DOWNLOAD WEIGHTS
READY
RESET ALL
TRAINING CHARTS
// PREDICTION RESULTS
CONFIDENCE SCORES
thumbs up 22.1%
peace 27.0%
yooo 50.9%
TOP PREDICTION
yooo
50.9% confidence
DATASET OVERVIEW
thumbs up 182 samples
peace 172 samples
yooo 184 samples
MEMORY MONITOR

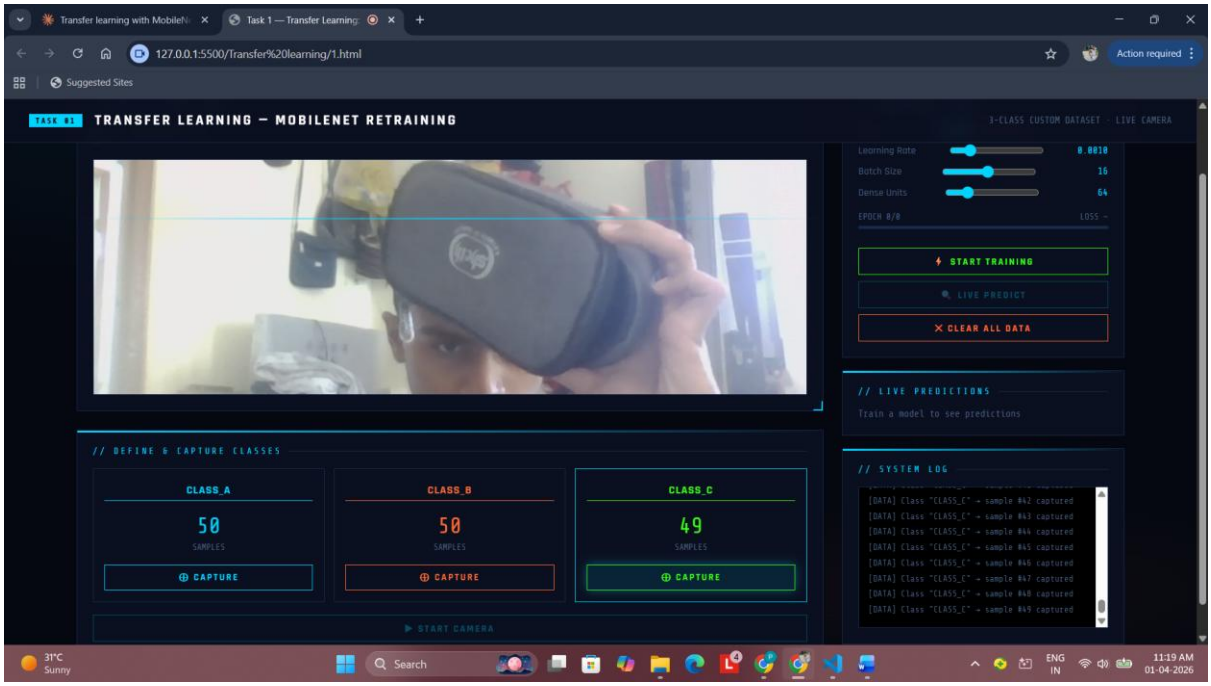

NeuralLens
TRANSFER LEARNING STUDIO
HOLD TO CAPTURE
HYPERPARAMETERS
LR
BATCH
0.0001
16
EPOCHS
50
PIPELINE
TRAIN MODEL
STOP PREDICTING
DOWNLOAD WEIGHTS
TRAINING STATS
Epoch
Loss
Accuracy
Samples 548
READY
RESET ALL
TRAINING CHARTS
// PREDICTION RESULTS
CONFIDENCE SCORES
thumbs up 18.7%
peace 58.0%
yooo 23.3%
TOP PREDICTION
peace
58.0% confidence
DATASET OVERVIEW
thumbs up 183 samples
peace 181 samples
yooo 184 samples
MEMORY MONITOR

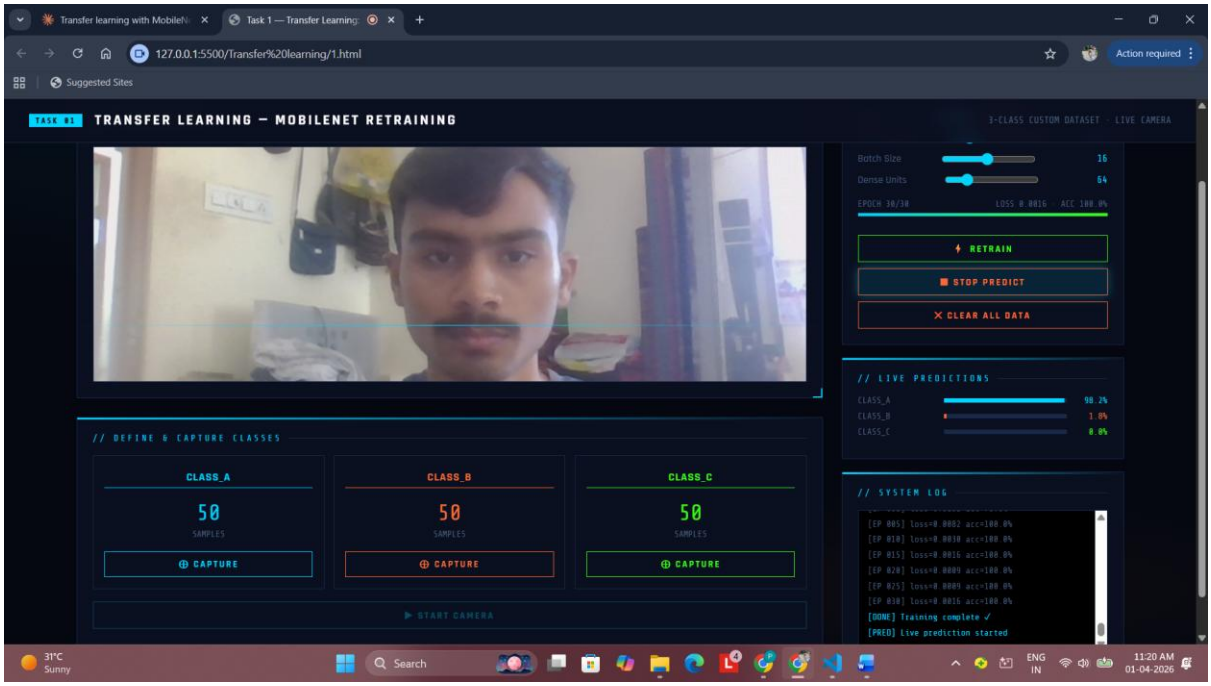

NeuralLens
TRANSFER LEARNING STUDIO
HYPERPARAMETERS
LR
BATCH
0.0001
16
EPOCHS
50
PIPELINE
TRAIN MODEL
STOP PREDICTING
DOWNLOAD WEIGHTS
TRAINING STATS
Epoch
Loss
Accuracy
Samples 550
READY
RESET ALL
TRAINING CHARTS
// PREDICTION RESULTS
CONFIDENCE SCORES
peace 37.4%
yooo 19.6%
thumbs up 43.0%
TOP PREDICTION
thumbs up
43.0% confidence
DATASET OVERVIEW
peace 181 samples
yooo 184 samples
thumbs up 185 samples
MEMORY MONITOR

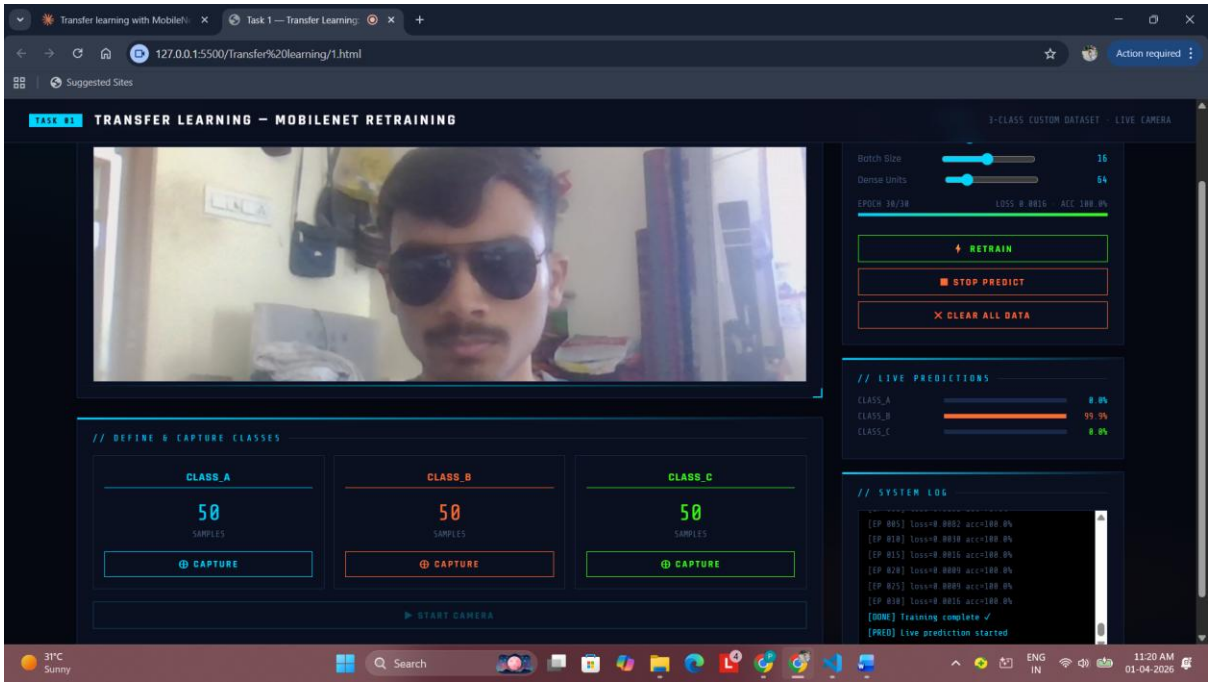
Assignment 1

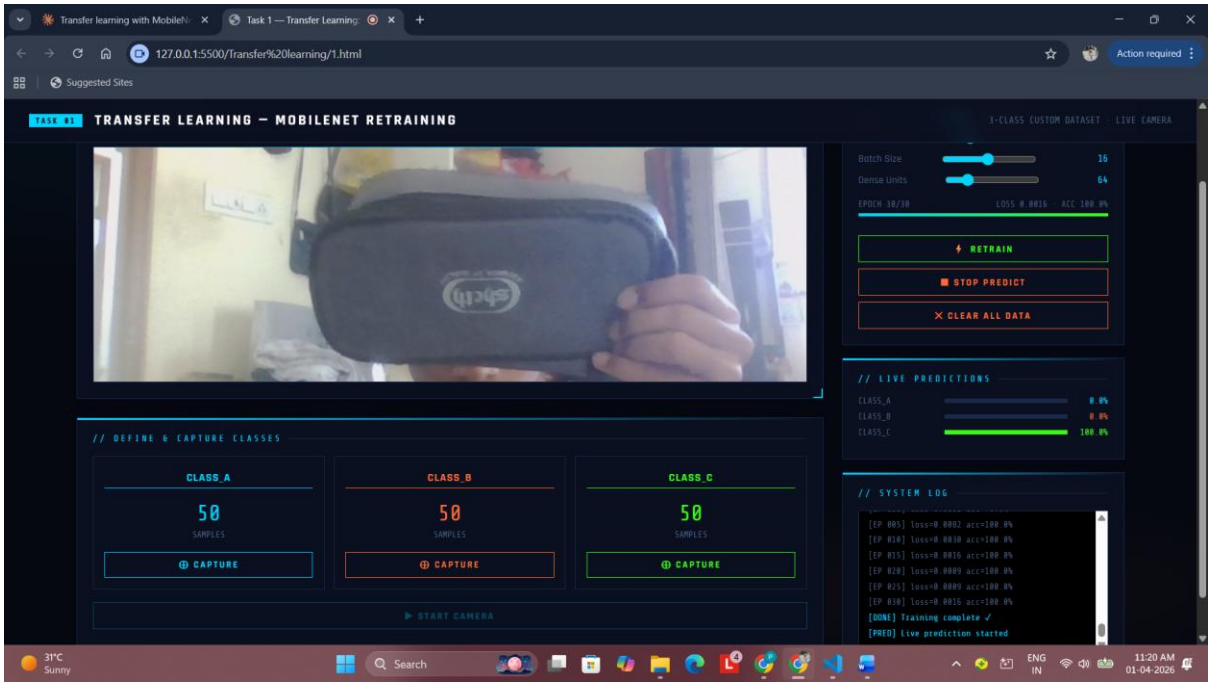




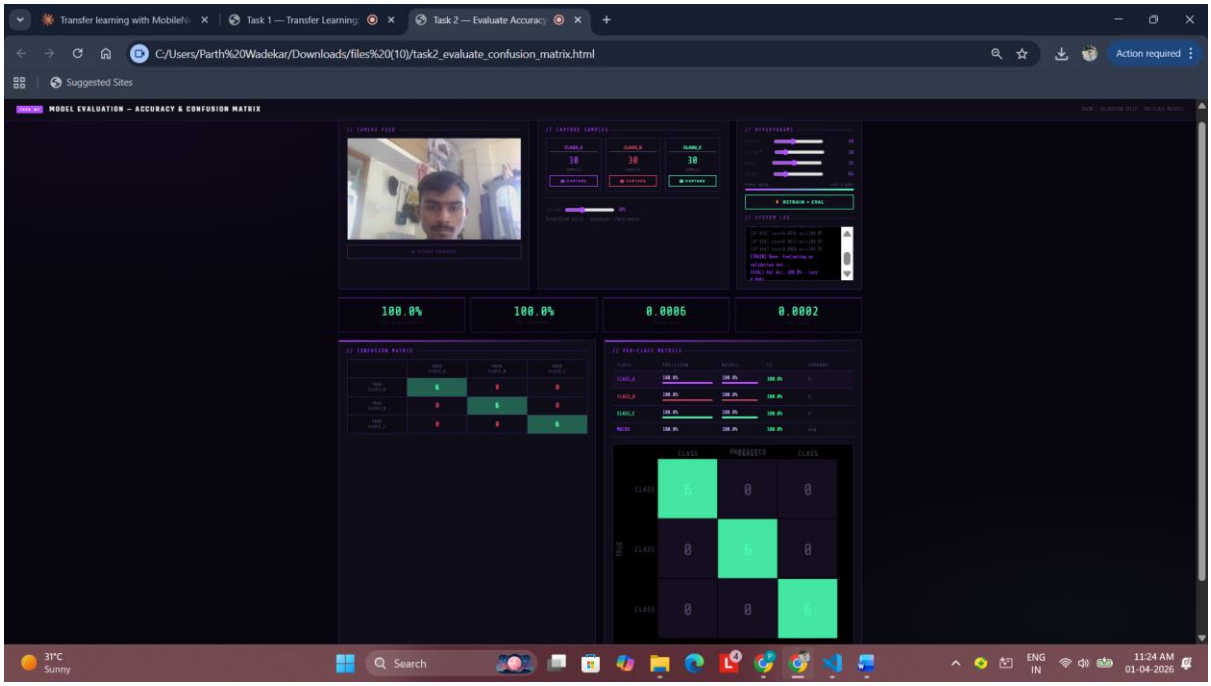

Transfer learning with MobileN
Task 1 — Transfer Learning
127.0.0.1:5500/Transfer%20learning/1.html
Action required
Suggested Sites
TASK 01
TRANSFER LEARNING — MOBILENET RETRAINING
3-CLASS CUSTOM DATASET · LIVE CAMERA
Learning Rate
0.0010
Batch Size
16
Dense Units
64
EPOCH 0/0
LOSS –
START TRAINING
LIVE PREDICT
CLEAR ALL DATA
// LIVE PREDICTIONS
Train a model to see predictions
// DEFINE & CAPTURE CLASSES
CLASS_A
50
SAMPLES
CAPTURE
CLASS_B
50
SAMPLES
CAPTURE
CLASS_C
49
SAMPLES
CAPTURE
START CAMERA
// SYSTEM LOG
[DATA] Class "CLASS_C" → sample #42 captured
[DATA] Class "CLASS_C" → sample #43 captured
[DATA] Class "CLASS_C" → sample #44 captured
[DATA] Class "CLASS_C" → sample #45 captured
[DATA] Class "CLASS_C" → sample #46 captured
[DATA] Class "CLASS_C" → sample #47 captured
[DATA] Class "CLASS_C" → sample #48 captured
[DATA] Class "CLASS_C" → sample #49 captured
31°C
Sunny
Search
ENG
IN
11:19 AM
01-04-2026


Transfer learning with MobileN
Task 1 — Transfer Learning
127.0.0.1:5500/Transfer%20learning/1.html
Action required
Suggested Sites
TASK 01
TRANSFER LEARNING — MOBILENET RETRAINING
3-CLASS CUSTOM DATASET · LIVE CAMERA
Batch Size
16
Dense Units
64
EPOCH 30/30
LOSS 0.0016 · ACC 100.0%
RETRAIN
STOP PREDICT
CLEAR ALL DATA
// LIVE PREDICTIONS
CLASS_A
98.2%
CLASS_B
1.8%
CLASS_C
0.0%
// DEFINE & CAPTURE CLASSES
CLASS_A
50
SAMPLES
CAPTURE
CLASS_B
50
SAMPLES
CAPTURE
CLASS_C
50
SAMPLES
CAPTURE
START CAMERA
// SYSTEM LOG
[EP 005] loss=0.0082 acc=100.0%
[EP 010] loss=0.0010 acc=100.0%
[EP 015] loss=0.0016 acc=100.0%
[EP 020] loss=0.0009 acc=100.0%
[EP 025] loss=0.0009 acc=100.0%
[EP 030] loss=0.0016 acc=100.0%
[DONE] Training complete ✓
[PRED] Live prediction started
31°C
Sunny
Search
ENG
IN
11:20 AM
01-04-2026

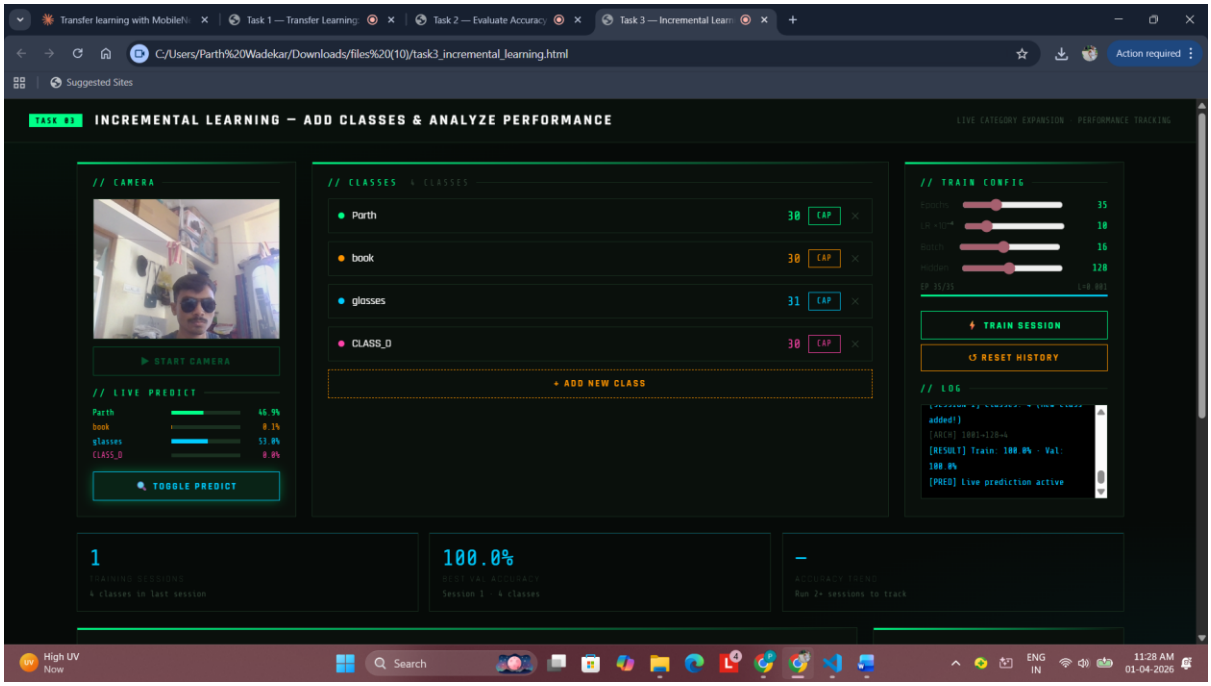

Transfer learning with MobileN
Task 1 — Transfer Learning
127.0.0.1:5500/Transfer%20learning/1.html
Action required
Suggested Sites
TASK 01
TRANSFER LEARNING — MOBILENET RETRAINING
3-CLASS CUSTOM DATASET · LIVE CAMERA
Batch Size
16
Dense Units
64
EPOCH 30/30
LOSS 0.0016 · ACC 100.0%
RETRAIN
STOP PREDICT
CLEAR ALL DATA
// LIVE PREDICTIONS
CLASS_A
0.0%
CLASS_B
99.9%
CLASS_C
0.0%
// DEFINE & CAPTURE CLASSES
CLASS_A
50
SAMPLES
CAPTURE
CLASS_B
50
SAMPLES
CAPTURE
CLASS_C
50
SAMPLES
CAPTURE
START CAMERA
// SYSTEM LOG
[EP 005] loss=0.0082 acc=100.0%
[EP 010] loss=0.0030 acc=100.0%
[EP 015] loss=0.0016 acc=100.0%
[EP 020] loss=0.0009 acc=100.0%
[EP 025] loss=0.0009 acc=100.0%
[EP 030] loss=0.0016 acc=100.0%
[DONE] Training complete ✓
[PRED] Live prediction started
31°C
Sunny
Search
ENG
IN
11:20 AM
01-04-2026

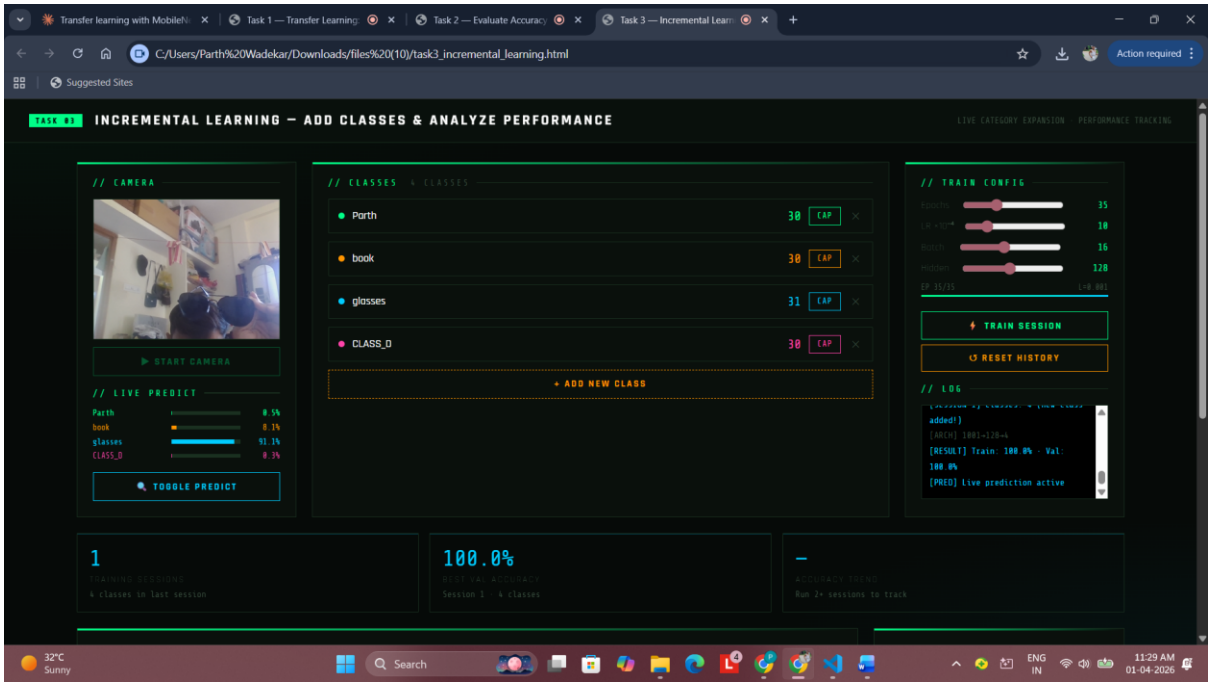

Transfer learning with MobileN
Task 1 — Transfer Learning
127.0.0.1:5500/Transfer%20learning/1.html
Action required
Suggested Sites
TASK 01
TRANSFER LEARNING — MOBILENET RETRAINING
3-CLASS CUSTOM DATASET · LIVE CAMERA
Batch Size
16
Dense Units
64
EPOCH 30/30
LOSS 0.0016 · ACC 100.0%
RETRAIN
STOP PREDICT
CLEAR ALL DATA
// LIVE PREDICTIONS
CLASS_A
0.0%
CLASS_B
0.0%
CLASS_C
100.0%
// DEFINE & CAPTURE CLASSES
CLASS_A
50
SAMPLES
CAPTURE
CLASS_B
50
SAMPLES
CAPTURE
CLASS_C
50
SAMPLES
CAPTURE
START CAMERA
// SYSTEM LOG
[EP 005] loss=0.0082 acc=100.0%
[EP 010] loss=0.0030 acc=100.0%
[EP 015] loss=0.0016 acc=100.0%
[EP 020] loss=0.0009 acc=100.0%
[EP 025] loss=0.0009 acc=100.0%
[EP 030] loss=0.0016 acc=100.0%
[DONE] Training complete ✓
[PRED] Live prediction started
31°C
Sunny
Search
ENG
IN
11:20 AM
01-04-2026

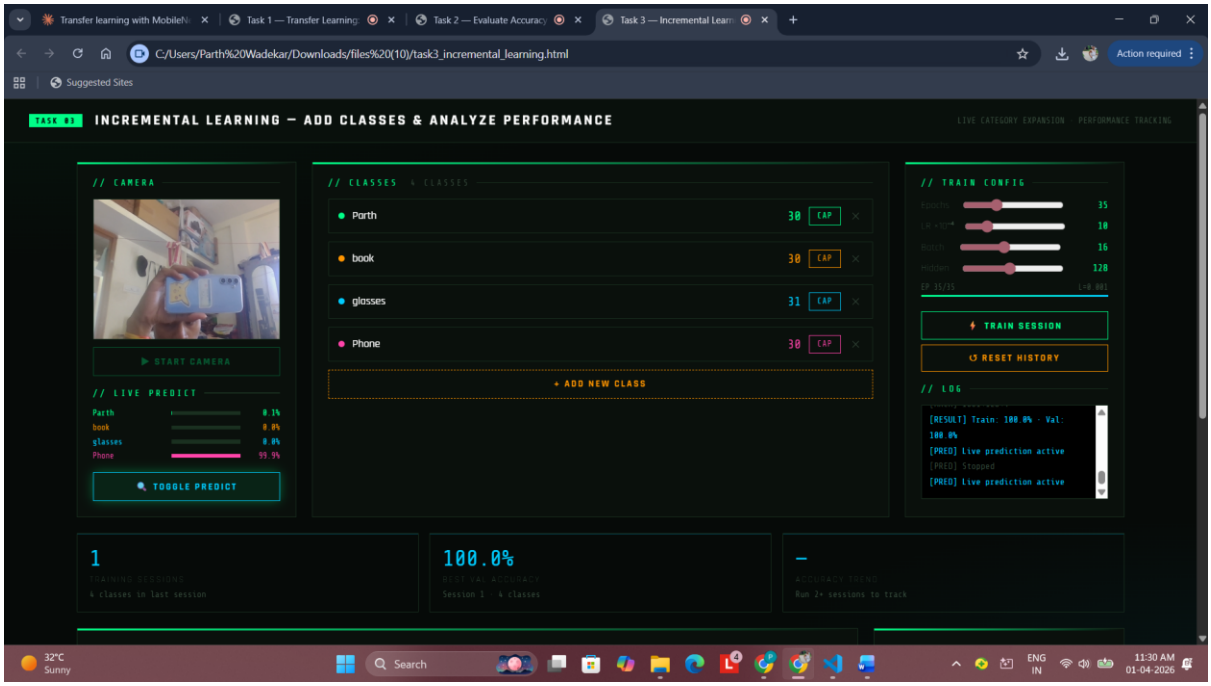
Assignment 2

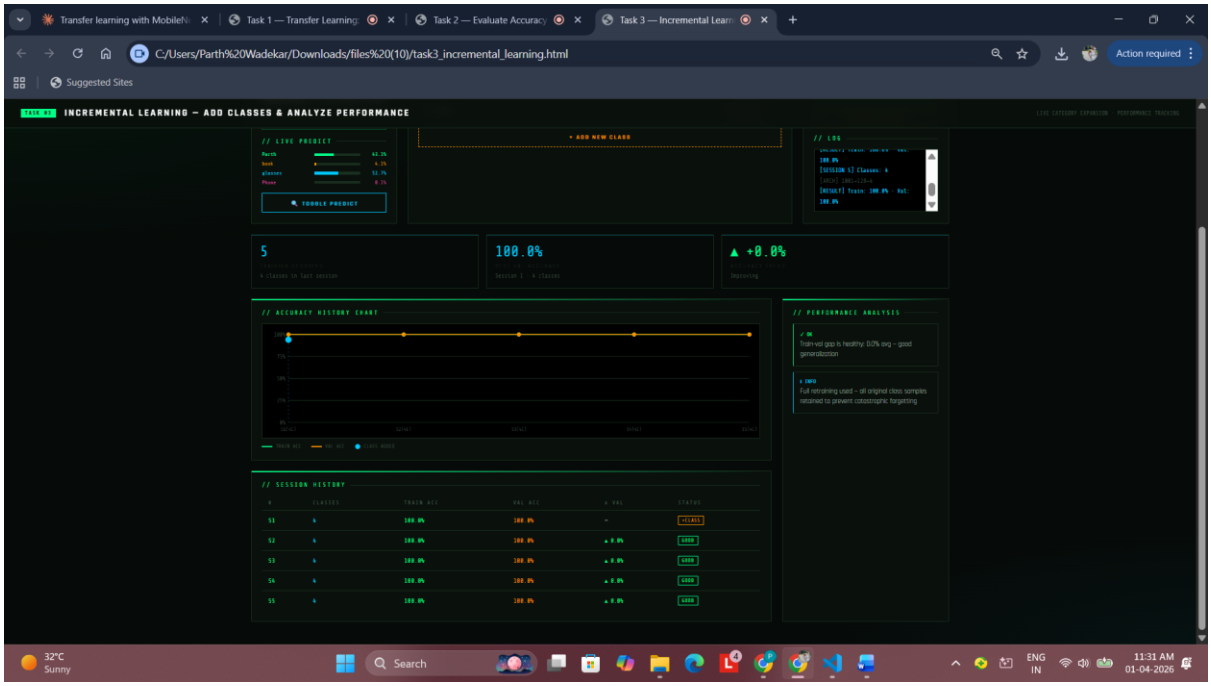


Assignment 3




Transfer learning with MobileN
Task 1 — Transfer Learning
Task 2 — Evaluate Accuracy
Task 3 — Incremental Learn
C:/Users/Parth%20Wadekar/Downloads/files%20(10)/task3_incremental_learning.html
Action required
Suggested Sites
TASK 03
INCREMENTAL LEARNING — ADD CLASSES & ANALYZE PERFORMANCE
LIVE CATEGORY EXPANSION · PERFORMANCE TRACKING
// CAMERA
START CAMERA
// LIVE PREDICT
Parth 0.5%
book 0.3%
glasses 91.3%
CLASS_D 0.3%
TOGGLE PREDICT
// CLASSES 4 CLASSES
Parth 30 CAP
book 30 CAP
glasses 31 CAP
CLASS_D 30 CAP
+ ADD NEW CLASS
// TRAIN CONFIG
Epochs 35
LR ×10⁻⁴ 10
Batch 16
Hidden 128
EP 35/35 L=0.001
TRAIN SESSION
RESET HISTORY
// LOG
added!)
[ARCH] 1001→128→4
[RESULT] Train: 100.0% · Val: 100.0%
[PRED] Live prediction active
1
TRAINING SESSIONS
4 classes in last session
100.0%
BEST VAL ACCURACY
Session 1 · 4 classes
—
ACCURACY TREND
Run 2+ sessions to track
32°C
Sunny
Search
ENG
IN
11:29 AM
01-04-2026


Transfer learning with MobileN
Task 1 — Transfer Learning
Task 2 — Evaluate Accuracy
Task 3 — Incremental Learn
C:/Users/Parth%20Wadekar/Downloads/files%20(10)/task3_incremental_learning.html
Action required
Suggested Sites
TASK 03
INCREMENTAL LEARNING — ADD CLASSES & ANALYZE PERFORMANCE
LIVE CATEGORY EXPANSION · PERFORMANCE TRACKING
// CAMERA
START CAMERA
// LIVE PREDICT
Parth 0.1%
book 0.0%
glasses 0.0%
Phone 99.9%
TOGGLE PREDICT
// CLASSES 4 CLASSES
Parth 30 CAP
book 30 CAP
glasses 31 CAP
Phone 30 CAP
+ ADD NEW CLASS
// TRAIN CONFIG
Epochs 35
LR ×10⁻⁴ 10
Batch 16
Hidden 128
EP 35/35 L=0.001
TRAIN SESSION
RESET HISTORY
// LOG
[RESULT] Train: 100.0% · Val: 100.0%
[PRED] Live prediction active
[PRED] Stopped
[PRED] Live prediction active
1
TRAINING SESSIONS
4 classes in last session
100.0%
BEST VAL ACCURACY
Session 1 · 4 classes
—
ACCURACY TREND
Run 2+ sessions to track
32°C
Sunny
Search
ENG
IN
11:30 AM
01-04-2026


Transfer learning with MobileNet
Task 1 — Transfer Learning
Task 2 — Evaluate Accuracy
Task 3 — Incremental Learn
C:/Users/Parth%20Wadekar/Downloads/files%20(10)/task3_incremental_learning.html
Action required
Suggested Sites
TASK 03
INCREMENTAL LEARNING — ADD CLASSES & ANALYZE PERFORMANCE
// LIVE PREDICT
Parth
book
glasses
Phone
TOGGLE PREDICT
+ ADD NEW CLASS
// LOG
[SESSION 5] Classes: 6
[RESULT] Train: 100.0% · Val: 100.0%
5
4 classes in last session
100.0%
Session 1 · 4 classes
▲ +0.0%
Improving
// ACCURACY HISTORY CHART
// PERFORMANCE ANALYSIS
✓ OK
Train-val gap is healthy: 0.0% avg – good generalization
ℹ INFO
Full retraining used – all original class samples retained to prevent catastrophic forgetting
// SESSION HISTORY
CLASSES TRAIN ACC VAL ACC Δ VAL STATUS
S1 4 100.0% 100.0% – +CLASS
S2 4 100.0% 100.0% ▲ 0.0% GOOD
S3 4 100.0% 100.0% ▲ 0.0% GOOD
S4 4 100.0% 100.0% ▲ 0.0% GOOD
S5 4 100.0% 100.0% ▲ 0.0% GOOD
32°C
Sunny
Search
ENG
IN
11:31 AM
01-04-2026